Snehal Yelkar

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Senior Full-Stack AI Engineer | Agentic AI · RAG · MCP · AWS Bedrock · React

Summary

Senior engineer with 7 years of experience, the last 2+ building a production multi-tenant AI platform on AWS: agentic assistants on Bedrock AgentCore, MCP tool servers, hybrid RAG search over 300K+ documents, and React micro-frontends.

Technical Skills

AI & Agents: Amazon Bedrock (Claude, Nova, Titan), Bedrock AgentCore, Strands Agents, LangChain, CrewAI, MCP, A2A, AG-UI, RAG, Hybrid Search, Cohere Rerank, Multimodal Embeddings, LLM Evaluation (DeepEval), Prompt Engineering
Languages: Python, TypeScript, JavaScript, SQL
Backend & Data: FastAPI, Pydantic, SQLAlchemy, Alembic, Node.js, REST, OAuth 2.1/JWT, Microservices, Event-Driven Architecture, Aurora PostgreSQL, pgvector, DynamoDB
Frontend: React, TypeScript, Vite, single-spa Micro-frontends, CopilotKit, Tailwind CSS, Playwright
Cloud & DevOps: AWS (Lambda, API Gateway, Step Functions, Kinesis, SNS, SQS, ECS, Cognito, CloudFront), Terraform, SAM, Docker, CI/CD, CloudWatch, OpenTelemetry, Datadog

Experience

Coforge (formerly Encora) — Pune, India

Senior Software Engineer – AI Platform — *Apr 2024 – Present*

- Core engineer on a multi-tenant enterprise AI platform (27 microservices, Python/FastAPI) delivering hybrid search, an agentic AI assistant and document intelligence inside a PLM product; **lead a team of engineers**
- Built an **agentic AI assistant**: Strands Agents on **Bedrock AgentCore** orchestrating MCP tool servers and A2A specialist agents, streamed to a React/CopilotKit micro-frontend over AG-UI
- Built an **OAuth 2.1 MCP gateway** exposing 6 MCP servers to agents with per-tenant, per-user access control
- Built an event-driven **RAG ingestion pipeline** (Kinesis, SQS, Step Functions) enriching and indexing 300K+ documents and images with Bedrock LLMs and multimodal embeddings, cutting manual tagging by ∼50%
- Built **hybrid search** on Aurora PostgreSQL + pgvector: keyword, semantic and visual lanes fused with weighted RRF and Cohere reranking; **cut cold first-query latency from 45s to 3–5s**
- Moved PLM API calls out of the search micro-frontend into a FastAPI **Backend-for-Frontend**, carrying the user's PLM session in the gateway JWT so the browser never holds PLM credentials
- Turned hard-coded similarity and roll-up logic into a **config-driven pipeline** with a React admin UI, live preview and AI-suggested configuration, matching legacy output row for row before cutover
- Built **self-service tenant onboarding** on AWS Service Catalog with selectable database tiers (Aurora Serverless or provisioned RDS) and Cognito app-client setup
- **Led the SAM to Terraform migration** (21 stacks) across dev, QA, preprod and prod in multiple AWS accounts and 2 regions; brought up production and built CI deploy-to-prod with a manual approval gate
- Own **production reliability (SRE)**: incident response and permanent root-cause fixes for intermittent auth failures from cached authorizer denials, 37K+ silently dropped records, and a Bedrock quota leak blocking tenant onboarding

Software Engineer – Full Stack — *Oct 2019 – Mar 2024*

- Led search migration to Elasticsearch, cutting query latency from 30s to 900ms across 1000+ daily searches
- Built React and TypeScript front-ends and Node.js REST APIs for large, complex datasets, with unit and integration tests (Jest, TDD)

Education & Certification

University of Pune (SPPU) — Pune, India

Bachelor of Engineering — *2015 – 2019*

AWS Certified Developer – Associate — *Amazon Web Services*